Eric Nguyen

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Education

University of California, Irvine — Irvine, CA
B.S. Computer Engineering (transferring Fall 2027) — *Expected June 2029*
- Orange Coast College, A.S. Computer Science | GPA: 3.72 | Kroll Award, Garrison Fellow, Student Leader of the Year

Work Experience

Product & Solutions Engineer — Midway City, CA
Plethora.so — Software Development Start-up — *April 2024 – Aug 2026*
- A meal-prep client hand-built menus, labels, and cut sheets every two weeks; shipped Chef Nourish (Next.js, TypeScript, PostgreSQL, Vercel) on a 3-engineer team, cutting the work from **5 hours to 30 minutes (90%)**.
- A mislabeled allergen could kill a customer, so designed a **7-category, 349-tag** system (**9,581 ingredient rules**): **0 leaks in 1,526 stress-tested dishes**.

Operations Assistant & Automation Developer — Costa Mesa, CA
Garrison Honors Center, Orange Coast College — *Jan 2024 – Present*
- Staff spent 30–45 minutes hand-checking each of 300+ honor-society applications per semester; shipped an offline desktop app (Python, PySide6, Pydantic, pyodbc/MS Access), **cutting each check to seconds**.
- Parsed HTML transcripts (Beautiful Soup) to check **148 requirements across 17 societies**, previewing every write to the registrar's 20-year-old Access database.

Projects

Luyten — Cal Hacks Sponsor Track Winner (SpacetimeDB) — Cal Hacks, UC Berkeley
TypeScript, SvelteKit, SpacetimeDB, Claude Agent SDK, Vercel AI SDK (Grok, Gemini) — *Oct 2026*
- Built the harness for **6 AI agents** to do due diligence and close a US/UK startup's books on CRS and Capital One data: 4 workers propose (books, revenue, credit, consolidation), a Red Team critiques, an Advisor rules.
- Implemented a SpacetimeDB database as the ledger: identity-checked reducers refuse any agent write outside its role, and one connection sustained **66K gated writes/s**, **3**× Postgres.
- Hit **100%** of expected credit decisions, sorting customers into **5 trust tiers**.

SLAP! — Live-Caption Classroom Device, Hardware & Software — Orange Coast College
TypeScript, Node (Hono, WebSockets), React Router, Postgres, Deepgram, ESP32-S3 — *Oct 2026*
- A lecture is gone once it's spoken, and worse with a language barrier, so built a slap-activated ESP32-S3 device with a teammate that streams class audio to a shared page: live captions in English, Vietnamese and Japanese.
- Cut caption delay **78% (555 to 121 ms)** by replacing Deepgram's SDK with a raw WebSocket, so captions feel live.
- Picked Google Cloud over 4 local translation models (**167 ms** vs 305+ ms); a bigger class costs no more to translate.
- Turned saved lectures into notes, flashcards and study plans; **won 2nd place, Education track**, OCC Hacks 2026.

OCC Hacks 2026 — Hackathon Website — Orange Coast College
Next.js, React, TypeScript, Tailwind, Supabase (Postgres, Auth, RLS), Resend, PostHog — *May 2026 – Oct 2026*
- Solo-built the site, 3 sign-up forms and the dashboard organizers ran the event from (QR check-in, emails, CSV export).
- Took **216 applications** in 6 weeks (**141 confirmed**), 38% in the final week; **12,267 page views** from ∼1,500 visitors.

LLM Math Reasoning — Fine-Tuning Research — Giles T. Brown Research Symposium
AI Research Fellow | LoRA fine-tuning, 4-bit quantization, dataset curation — *Jan 2025 – April 2025*
- Reasoning models waste compute on filler steps, so tested whether shorter reasoning makes a small model better at math: with a teammate, built **137 verified math solutions** (GSM8K, AIME, Putnam), condensed by a larger model.
- LoRA fine-tuned DeepSeek-R1-Distill-Qwen-7B on one RTX 4060 Ti, **beating the base model on every exam**: unseen AIME 2024 **53.3% vs 46.7% pass@1 (+14%)**, **66.6% vs 56.6% pass@3 (+18%)**; AMATYC **+15.8%**.

Leadership

Founder & President — Costa Mesa, CA
Iota Xi, Orange Coast College — *Feb 2024 – May 2026*
- Founded OCC's first CS honor society (**100+ members** in first semester).

Lead Organizer — Costa Mesa, CA
OCC Hacks 2026, Spookathon 2025, HackCC 2024 — *Feb 2024 – Oct 2026*
- Led a 6-department team to run 3 hackathons for 100+ attendees each; **raised $31,000+** and won OCC's Event of the Year.

Skills

Languages: TypeScript, JavaScript, Python, C++, SQL, HTML/CSS
Frontend: React, Next.js, React Router, Tailwind CSS, shadcn/ui
Backend & Tools: Node.js, Hono, WebSockets, PostgreSQL, Supabase, Neon, Git, Docker, Vercel, PostHog, ESP32-S3